Sign memory amplifies counting noise in market order flow

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Counting statistics can separate broad transfer sizes from temporal organisation in a market. We analyse both raw signed volume and volume-normalised imbalance in four primary cryptocurrency assets and a post-hoc four-asset extension. A stratified permutation test first destroys the complete signed-flow ordering and then separately destroys sign or magnitude ordering. Moving-block bootstrap intervals account for uncertainty in weekly variance. For both flow definitions, preserving the sign path recovers substantially more of the observed variance than preserving the magnitude path: sign persistence is the dominant temporal component, while magnitude clustering and sign–magnitude coupling remain measurable. The result is independent of a transfer-size quantum and survives calendar-conditioning checks. We also show why two common supports are not independent discoveries: accumulated-flow variance is fixed by its autocorrelation, and a linear fluctuation symmetry has an exact Gaussian null. The novel content is the attribution of memory-amplified counting noise to temporal sign organisation.

Introduction. A market has two sides, and the net signed flow of executed orders between them is a current: it counts something transferred, it fluctuates, and it responds to what drives it. This reading makes available a measurement programme built for transport in open, driven systems, whose most informative layer is the full counting statistics of the charge transferred in a finite window [1–4]. Moments of the signed flow accumulated over a window have been measured before, and their anomalous scaling with window length attributed to order splitting [5]; what has not been carried over is the counting framework itself — the normalisation of the noise to its Poisson value through an effective transfer quantum, and the reading of the log-ratio slope as an affinity conjugate to a current.

There is a reason to be careful in doing so. Statistical-physics readings of markets have proliferated: effective temperatures have been defined from return ratios [6], from order-book energies [7], and from holding-time asymmetries [8], and a fluctuation–dissipation relation has been reported for an effective Brownian particle in the book [9]. Several of these rest on observing a log-ratio of forward and reversed distributions that is linear in the observable. We show first that this linearity, by itself, establishes nothing — and then use that fact to isolate what does.

Setup. The primary panel uses hourly and four-hourly candles for BTC, ETH, BNB, and SOL against USDT over four years — eight series, with 3.5×10^4 hourly observations per asset. A post-hoc robustness panel adds hourly XRP, ADA, DOGE, and AVAX data to the weekly-null analysis; it is kept separate from the original decision rule. Public candle data carry total base volume V_t and taker-buy base volume V_t^+, so no proprietary feed is required. We analyse in parallel raw signed volume $\varepsilon_t^{(r)} = 2V_t^+ - V_t$ and dimensionless imbalance $\varepsilon_t^{(n)} = \varepsilon_t^{(r)}/V_t$ (zero when $V_t = 0$). This resolves a definition that earlier drafts used inconsistently between counting and correlation analyses. Figures and quoted exponents are for hourly data unless stated otherwise. Let $Q_T = \sum_{t=1}^{T} \varepsilon_t$ be the declared flow accumulated in a window of T candles, and

$$s(Q) = \ln \frac{P(Q_T = +Q)}{P(Q_T = -Q)} \tag{1}$$

its fluctuation symmetry function. Analysis code, deposited numerical outputs, and data provenance are available [10]; the measurement and model are developed at length in a companion article [11].

The identity. For a Gaussian Q_T with mean m_T and variance V_T,

$$s(Q) = \frac{(Q + m_T)^2 - (Q - m_T)^2}{2V_T} = \frac{2m_T}{V_T}\, Q, \tag{2}$$

exactly and always. Writing the cumulant generating function $F_T(\chi) = \ln \left\langle e^{\chi Q_T} \right\rangle = \chi m_T + \frac{1}{2}\chi^2 V_T$ and substituting $\chi \to -A_T - \chi$ with $A_T = 2m_T/V_T$ returns $F_T(\chi)$ identically, so the Gallavotti–Cohen symmetry [12, 13] is satisfied as well.

The consequence is a methodological guardrail. *A measured linear fluctuation symmetry whose slope equals* $2\langle Q \rangle / \mathrm{Var}(Q)$ *is not, by itself, evidence of a nontrivial fluctuation theorem.* The measured affinity tracks this Gaussian slope across windows and assets. Cubic fits to $s(Q)$ and independently measured higher cumulants are reported in the Supplemental Material; the latter do not converge uniformly to zero with scale, so we neither impose Gaussianity nor define a Gaussian crossover scale [14]. Agreement with Eq. (2) therefore does not distinguish a nontrivial symmetry from its Gaussian null explanation.

Nor do we establish a current large-deviation principle. Its conventional speed is T, while the measured variance grows as T^{ν} with $\nu > 1$; an alternative speed and rate function would have to be demonstrated. We therefore use $s(Q)$ only as a finite-window distributional diagnostic, not as evidence for an entropy-production theorem.

Memory amplification beyond the marginal. Because trades have a broad size distribution, the absolute Fano factor depends on the chosen transfer quantum,

$$\mathcal{F}(T) = \frac{\mathrm{Var}(Q_T)}{\bar{q}^2 \langle N_T \rangle}, \tag{3}$$

where N_T is the number of trades. Hourly candles do not contain individual sizes: the earlier estimator $\bar{q}_{\mathrm{proxy}} = \mathrm{median}_t(V_t/N_t)$ is a median of hourly *mean* sizes. Public trade files for one complete UTC day contain 9.80 million BTCUSDT and 4.97 million ETHUSDT operations. Aggregating them reproduces hourly volume, taker-buy volume, and trade count to machine precision. Their exact median sizes are smaller than $\bar{q}_{\mathrm{proxy}}$ by factors 87.5 and 71.6, respectively. Thus the previous absolute $\mathcal{F}$ values were conservative, but their magnitude is normalisation-sensitive; moreover, even an independent compound-Poisson process has $\mathcal{F} = \langle q^2 \rangle / \bar{q}^2$ and can be large without memory.

We therefore test memory with a variance ratio in which $\bar{q}$ cancels. We retain complete Monday–Sunday UTC weeks, remove the mean in every quarter–hour-of-week stratum, and permute within those strata. The joint null destroys the ordering of the complete signed-flow values. Two attribution nulls then act on the demeaned flow $\varepsilon_t = s_t m_t$: the sign null keeps the magnitude path m_t and permutes s_t, whereas the magnitude null keeps the sign path and permutes m_t. Every surrogate is re-centred in the same strata. The latter two nulls intentionally break contemporaneous sign–magnitude pairing and are attribution diagnostics, not generative market models.

With 9999 primary permutations, the joint-null ratio is 2.2–3.4 for raw signed volume and 3.0–4.3 for normalised imbalance [Fig. 1]. Destroying only sign order leaves nearly the same ratio, whereas destroying magnitude order lowers it to 1.7–2.2 (raw) and 1.5–1.9 (normalised). Thus the preserved sign path recovers substantially more weekly variance than the preserved magnitude path: sign persistence is the dominant component, although magnitude organisation and its coupling to sign remain non-negligible. Circular moving-block bootstrap intervals with 4–26-week blocks remain above unity for the joint comparison; the eight-week intervals are shown in Fig. 1.

The original predeclared rule required $V_{\mathrm{obs}} > 2Q_{0.9875}(V_{\mathrm{null}})$ for every asset under the raw-flow quarterly joint null. It failed because BNB did not pass. We report that failure without substituting the later normalised-flow result. Monthly and biweekly conditioning quantify scale sensitivity, and cross-asset weekly correlations show that the four assets are not independent replications [14].

The post-hoc extension reproduces the attribution pattern in all four added assets. Its joint-null ratios span 2.08–5.67 for raw flow and 2.83–5.64 for normalised flow; in every case, preserving the sign path recovers more variance than preserving the magnitude path. These observations strengthen generality but are not added to the predeclared gate.

For scale, we also compute a single-level, two-terminal noninteracting device with nonequilibrium Green functions [15] and standard counting fields [2, 16]. Under its conventional normalisation it yields $\mathcal{F} = 0.332$–0.341 and a variance exponent 0.94–0.99. This is a calibrated reference for short-memory fermionic transport, not a universal bound on a market statistic with heterogeneous classical transfer sizes. The exclusion in the data comes instead from the size-preserving temporal null.

A linked second-order check. The accumulated-flow variance and the autocorrelation are connected by an exact finite-sample identity, so they are not independent evidence. They do, however, expose whether a proposed scaling summary is internally adequate. For a single power-law kernel $C_\varepsilon(\tau) = A\tau^{-a}$ with $0 < a < 1$,

$$V_T = \sum_{t,t'=1}^{T} C_\varepsilon(t - t') \simeq \frac{2A}{(1-a)(2-a)}\, T^{2-a}, \tag{4}$$

so the counting variance grows with exponent $\nu = 2 - a$. At $a = 1$ the leading form is $T \log T$, and for $a > 1$ the correlation is summable and the asymptotic growth is diffusive. The measured C_ε, however, is not a single power law: fitting local slopes separately gives a fast regime a_f over $\tau \in [2, 10]$ and a slow one a_s over $\tau \in [10, 50]$, with $a_f > a_s$ in all four assets, and removing the daily seasonal component leaves both essentially unchanged. For the reference asset, $a_f = 1.00 \pm 0.15$ and $a_s = 0.17 \pm 0.13$.

Equation (4) then cannot hold with a single ν. Short windows sample the fast regime and long windows the slow one, so the *local* logarithmic slope must drift,

$$\nu_{\mathrm{loc}}(T):\; 2 - a_f \;\longrightarrow\; 2 - a_s, \tag{5}$$

here a band $[1.00, 1.83]$.

The broad-window single-power prediction fails in all eight asset–resolution series: predicted exponents are 1.45–1.88, while block estimates are 1.22–1.37. Mapping the two fitted regimes with the correct domain gives asset-specific point-estimate bands $[1.00, 1.83]$, $[1.13, 1.78]$, $[1.54, 1.71]$, and $[1.00, 1.84]$ for BTC, ETH, BNB, and SOL. BNB's measured 1.37 lies outside its band. Because amplitudes and crossover uncertainty were not fitted, these intervals are only consistency bands; they are not parameter-free predictions. What remains robust is global superdiffusion: $\nu = 1.31$–1.43 from block variance and 1.10–1.36 from three drift-resistant estimators, versus $\nu = 1$ for a short-memory reference.

The apparent *drift* of the local exponent is not identified. A drifting block mean adds a T^2 term to V_T,

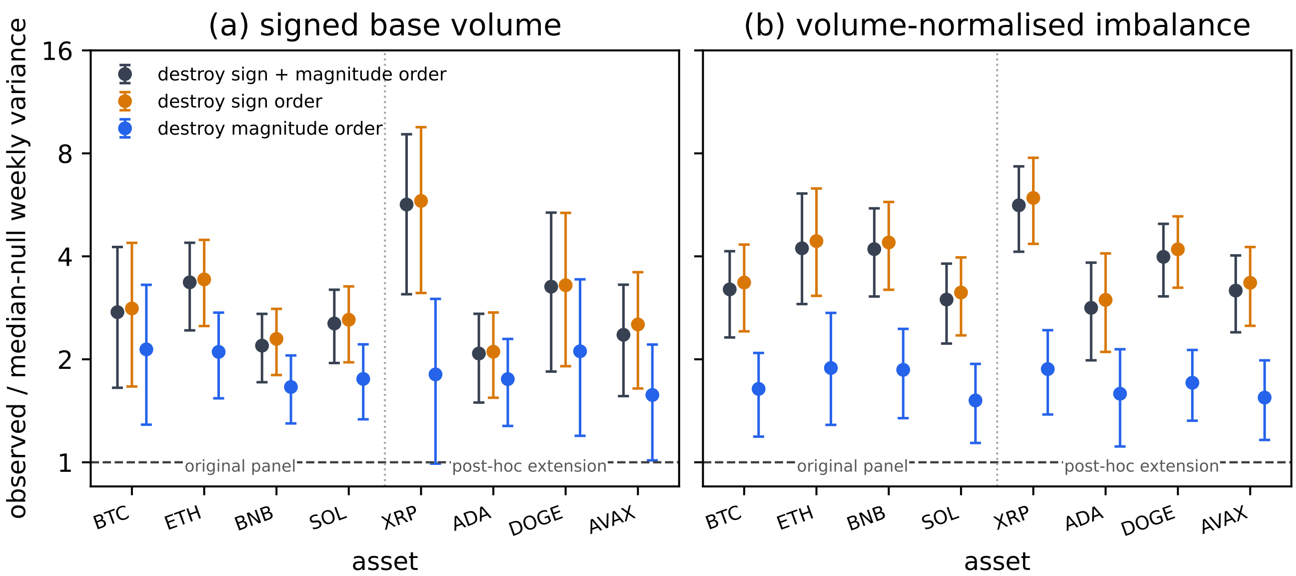

FIG. 1. Weekly variance relative to each stratified null for (a) raw signed volume and (b) normalised imbalance. Error bars are eight-week moving-block bootstrap 95% intervals conditional on the null median. Preserving sign order (blue) recovers more variance than preserving magnitude order (orange).

steeper than the kernel allows. The Haar (db1) difference between adjacent blocks separates the two, since $\mathrm{Var}(Q_T^{(2)} - Q_T^{(1)}) = 4V_T - V_{2T} = A(4 - 2^\nu)T^\nu$ carries the same exponent with a constant mean cancelled exactly. Applied directly to the flow, db2 additionally removes linear drift and db3 quadratic drift [17]. All four estimators agree to 0.01 on synthetic series of known ν, and the injected drift moves only the block one — so the comparison has the required sensitivity. On the data the block estimator is the highest of the four in every asset, by 0.13 to 0.29; injecting each asset's own measured drift onto a clean synthetic series reproduces up to 0.13 of that gap, and removing the drift by hand closes it. The drift-blind estimators neither reproduce the crossover nor exclude it, their local slopes carrying errors of ± 0.05 to ± 0.22 against a claimed excursion of 0.28. We therefore report Eq. (5) as consistent with the data and not discriminated by them, and make no inference from it against a one-component bath. The db1–db3 values and local slopes are tabulated in the Supplemental Material [14].

A corollary of Eq. (2) combined with Eq. (4) is that the affinity must dilute with scale as $A_T \propto T^{a-1}$; the parameter-free form $A_{T_2}/A_{T_1} = (T_2/T_1)(V_{T_1}/V_{T_2})$ reproduces the measured ratio to within 0.5–13% for the two assets whose affinity is stable across the sample, and fails badly for the two whose affinity sits at the noise level — failing, that is, precisely where the predicted quantity is not determined.

Discussion. The absolute Fano level is not a memory diagnostic because it depends on transfer sizes and the declared quantum. The normalisation-free result is instead that the weekly variance exceeds temporally disordered nulls under both flow definitions, with the sign path accounting for the larger share. The original all-assets stringent gate nevertheless failed, and the primary assets are correlated rather than independent replications. A separate four-asset extension reproduces the sign-dominance ordering. Superdiffusion is a consistent second-order description of the same memory, not a separate confirmation.

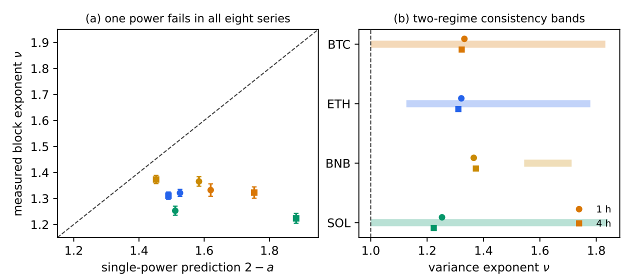

FIG. 2. Scaling reassessment. (a) The broad-window single-power prediction misses all eight series. Circles and squares denote hourly and four-hourly data. (b) Asset-specific two-regime consistency bands; BNB lies outside. Amplitudes and crossover uncertainty are not included, so the bands are not quantitative predictions.

The methodological point is equally important. A linear forward–reverse log-ratio is guaranteed for any Gaussian observable, so its appearance alone is not a discovery and its slope need not be independent. The informative content lies instead in normalisation-free null comparisons, higher cumulants, and the scale dependence of memory.

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